

EMA705 - Advanced Topics in Finite Elements

HW 3

Given: 20 April, 2026, due: **3 May, 2026**

Submit on Canvas.

Problem (30 points)

Assuming a St. Venant-Kirchhoff strain energy density function, $\sigma_{ij} = 2\mu\varepsilon_{ij} + \lambda\delta_{ij}\varepsilon_{kk}$, solve the following 2D boundary value problems using an **infinitesimal strain** mechanics FE implementation with a plane strain assumption. Use $\lambda = 9.15 \times 10^{12}$ Pa and $\mu = 1.83 \times 10^{10}$ Pa. This gives a **nearly incompressible** material ($\nu=0.499$).

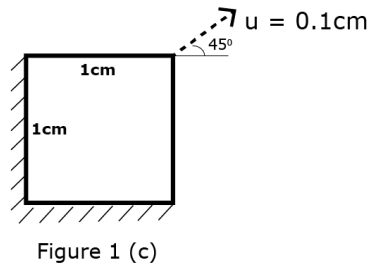
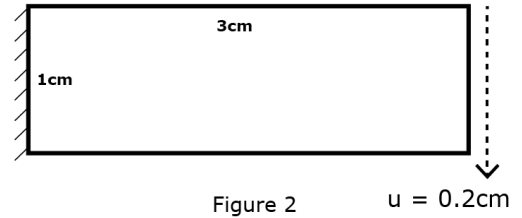
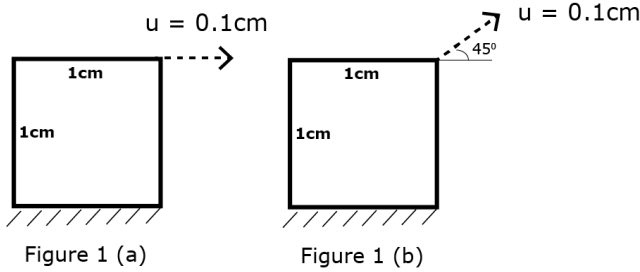
- [Part 1] Using classical **1-field** formulation of elasticity, and a square geometry ($1\text{cm} \times 1\text{cm}$) subjected to the boundary conditions shown in subfigures of Figure 1, with:
 - [Part 1(a)]: 2 *Linear Triangle* elements with one point quadrature. Use boundary conditions in Figure 1(a) and Figure 1(b).
 - [Part 1(b)]: 16 (4x4) *Linear Quadrilateral* elements with 1x1 point quadrature. Use boundary conditions in Figure 1(c).
 - [Part 1(c)]: 16 (4x4) *Quadratic Quadrilateral* elements with 2x2 point quadrature. Use boundary conditions in Figure 1(c).
- [Part 2] Using **2-field** formulation of incompressible elasticity, and a square geometry ($1\text{cm} \times 1\text{cm}$) subjected to the boundary conditions shown in subfigures of Figure 1, with:
 - [Part 2(a)]: 2 *Linear Triangle* elements with optimal pressure DOFs and appropriate quadrature. Use boundary conditions in Figure 1(a) and Figure 1(b).
 - [Part 2(b)]: 16 (4x4) *Linear Quadrilateral* elements with optimal pressure DOFs and appropriate quadrature. Use boundary conditions in Figure 1(c).
 - [Part 2(c)]: 16 (4x4) *Quadratic Quadrilateral* elements with optimal pressure DOFs and appropriate quadrature. Use boundary conditions in Figure 1(c).

Indicate the number of pressure DOFs used and the quadrature rule used in each case next to your displacement plots.

- [Part 3] Using **2-field** formulation of incompressible elasticity, and a beam geometry ($3\text{cm} \times 1\text{cm}$) subjected to the boundary conditions shown in Figure 2, with:
 - [Part 3(a)]: 12 *Linear Triangle* elements with optimal pressure DOFs and appropriate quadrature.
 - [Part 3(b)]: 24 (6x4) *Linear Quadrilateral* elements with optimal pressure DOFs and appropriate quadrature.
 - [Part 3(c)]: 24 (6x4) *Quadratic Quadrilateral* elements with optimal pressure DOFs and appropriate quadrature.

Indicate the number of pressure DOFs used and the quadrature rule used in each case next to your displacement plots.

Assume zero traction Neumann boundary conditions on boundaries where no boundary conditions explicitly specified.



Submit the source code and a single PDF containing the following plots on Canvas:

- Source code for the 2D FE program. The code should be well commented to clearly show the various steps of the FE solution procedure, and the boundary conditions and initial conditions.
- For each part, show the displacement plot of $\mathbf{u}(x)$ over the deformed configuration (deformed mesh) showing the deformed geometry and the displacement contours.